

[91]

SARDAR PATEL UNIVERSITY
Master of Science (Information Technology)-III Semester (NC)
PS03CINT53 : Computer Graphics and Multimedia
9th April, 2025, Wednesday

**Time: 2.00 PM To 5.00 PM****Marks: 70****Q1. Choose the most appropriate option for the following:****[8]**

- i. GKS stands for ____
a) Graphical Kernel System b) Graphical Kiosk System
c) Geographical Kernel System d) Geographical Kiosk System
- ii. The maximum number of points that can be displayed without overlap on a CRT is referred to as ____.
a) Pixel b) Resolution c) Aspect ratio d) None of given
- iii. Picture definition is stored in memory area called the ____.
a) Retrace buffer b) Re-enter buffer c) Refresh buffer d) Bounded buffer
- iv. Which of the following is NOT a type of font?
a) Bitmap b) Pixmap c) Outline d) None of given
- v. The physical distance between two actual picture elements (Actual Pixels) is known as ____
a) Dot pitch b) PIXDIT c) PIXDIST d) None of given
- vi. The ____ file format allows to store an animation sequence.
a) JPG b) GIF c) PNG d) None of given
- vii. The ____ color model is used in Color monitor.
a) YCbCr b) RGB c) CMY d) None of given
- viii. The process of removing noise from the audio signals is known as ____.
a) Masking b) Dithering c) Splicing d) None of given

Q2. Answer in short (Attempt any SEVEN)**[14]**

- i. Define Computer Graphics.
- ii. Explain Point clipping algorithm.
- iii. What do you mean by Shear transformation?
- iv. What is Blobby Object?
- v. List out the different facets of Multimedia.
- vi. List out the features of PNG graphics file format.
- vii. Define: Graphics file format and Color model.
- viii. Why there is need of Video in Multimedia project?
- ix. How many bits are required to store 5 seconds of audio file having 32 bit sound system.

(1)

[P.T.O.]

Q3. Do as directed:

- A What do you mean by output primitives? List out various output primitives with their attributes. [6]
B Explain briefly the steps to generate line using Bresenham's approach. [6]

OR

- B List out and explain briefly the Text Clipping algorithms. [6]

Q4. Do as directed:

- A Write short note on 3-D transformations. Explain any one of them in detail. [6]
B Draw and explain the 3-D viewing pipeline. [6]

OR

- B Discuss briefly the 3-dimensional display methods: Parallel projection and Perspective projection. [6]

Q5. Do as directed:

- A Define: Image digitization. Write short note on components of Image processing system. [6]
B Define: Sampling, Quantization, Image Compression, Image Processing, Pseudo coloring and Image Analysis. [6]

OR

- B Smooth the pixel $g(2,2)$ which is underlined, of a following grayscale image having 25 pixels, in 3×3 domain using median filter and mode filter. [6]

1	2	2	4	5
4	5	0	1	3
3	4	<u>8</u>	4	0
2	1	4	6	1
3	5	2	5	2

Q6. Do as directed:

- A Define: Multimedia, Digital Audio, Dithering, ADC process, Digitizing, Animation [6]
B Write short note on Audio cards. [6]

OR

- B Explain briefly the steps to create 3-Dimensional animation. [6]

